

Roll No.

TCE-302/TPE-304

B. TECH. (CE/PE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
BASIC SURVEYING/GEOMATIC ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Differentiate between plane and geodetic surveying. Give a classification of surveys based on the instruments used. (CO1)

OR

- (b) Describe the various source and types of errors in surveying measurement. (CO1)

2. (a) The area of a plot on a map is found, by planimeter, to be 40.55 cm². The scale of the map was 1 : 45000, but at present it is shrunk such that a line originally 8 cm on the map is now 7.5 cm. What is the correct field area in hectares ? (CO1)

OR

- (b) Define the following terms : (CO1)
(i) Main station

P. T. O.

- (ii) Tie station
- (iii) Main survey line
- (iv) Tie line
- (v) Base line
- (vi) Check line
- (vii) Offset
- (viii) Ranging
- (ix) Scale of map
- (x) Least count

3. (a) Describe different types of tapes commonly used in surveying stating the advantages of each. (CO1)

OR

- (b) A 20 m chain was found to be 10 cm too long after chaining distance of 2000 m. It was found to be 15 cm too long at the end of day's work after chaining a total distance of 3400 m. Find the true distance if the chain was correct before the commencement of the work. (CO1)

4. (a) The following bearings were taken in a closed traverse ABCD :

Line	Fore Bearing	Back Bearing
AB	65°15'	245°15'
BC	120°15'	300°15'
CD	185°00'	5°00'
DA	275°30'	95°30'

Calculate the interior angles of the traverse.

(CO2)

(3)

OR

- (b) Explain a bearing. What are the different systems of designation of bearings ? Explain each system with a neat sketch. (CO2)
5. (a) What are the advantages and disadvantages of method of repetition ? Compare method of repetition and reiteration. (CO2)

OR

- (b) Describe the different errors in theodolite work. How are they eliminated ? (CO2)